

VISION-LANGUAGE INTENT GROUNDING PIPELINE

Converting Natural Language Commands to 3D Affordance Bounding Primitives with Metric Leases

1. NATURAL LANGUAGE PROMPT

User: "Pick up the red thermal vial from shelf B"

2. MULTI-MODAL EMBEDDING

Cross-attention over RGB image tokens + prompt tokens

3. 3D SPATIAL GROUNDING

Projects 2D attention peak into calibrated 3D world frame

4. INTENT LEASE EMISSION

Emits 3D Goal Primitive + Expiring Lease ($t_{\text{expire}} = 2.0$ s)

5. DOWNSTREAM EXECUTION

Action chunk planner & 1 kHz MCU safety filter